

EPH 1100 - ANALYTICAL TOOLS FOR ECONOMICS AND POLITICAL SCIENCE

(3.0 credits) Fall 2026 - Syllabus



Schedule: Tuesday / Thursday – 10:00-11:15 AM
Location: Scarbrough Building: Tocqueville - Classroom 202
Instructor(s): JP Bastos
Email: jbastos@uaustin.org
Office Hours: TBD

Course Description

Legacy Description: This course introduces students to the most useful mathematical tools that will be applied in the remaining courses in economics and political science offered by the Center for Economics, Politics and History. Students learn each mathematical tool and then how to apply it to an economic or political problem. Students will become familiar with the meaning and use of the following key concepts: functions, utility maximization under certainty and uncertainty, solving a system of equations, using derivatives in optimization problem solving, probability and the main families of probability distribution functions, conditional and unconditional probability, the law of large numbers, the central limit theorem, variance, covariance, correlation, projection, regression, stochastic processes, cooperative and noncooperative games, and Nash Equilibrium. Students will also learn the basics of programming in R.

Course Objectives & Learning Outcomes

- Familiarize with the basic mathematical tools used to model and analyze problems in economics and political science, focusing on calculus, optimization, and linear algebra.
- Develop the problem-solving skills needed to set up and solve the canonical problems in economics for which these tools are required.
- Build a rigorous analytical foundation required to read and engage with theoretical and empirical work in subsequent courses and in the literature.
- Apply these tools to concrete economic and political problems, learning to translate real-world questions into formal models and interpret the results.

Assigned Texts

The primary reading for this course is the class notes, one set per class session, following the schedule below. You are expected to read the notes before each class, work through as many practice problems as you can, and come to class prepared to ask questions about anything you were not able to solve. There is no substitute for attempting the problems yourself first. **Any feedback on these notes is sincerely appreciated.**

In each class note, I reference the relevant chapters in [Hoy et al. \(2022\)](#), but is *optional*. It covers more than we will need, so use it as a reference if you want a more thorough or alternative treatment of any topic.¹

Grading Policy

Course grades are on a 100-point scale. All professors are required to target a course average of 85/100. See the UATX Academic Catalog for complete grading policy details.

Grades will be weighted as follows:

- **10% Class Participation.** Students will be called on during class to answer questions, come to the board, and to engage. You will be called on randomly, so come to class prepared.
- **10% Homework** (*pass/fail*). You are encouraged to work together. Homework will not be graded for correctness, only for completion. Late homework will be accepted but penalized.
- **15% Quizzes.** Written quizzes will be given at the start of a handful of random classes as a means of assessing your understanding of the readings and prior seminars.
- **30% Mid-term exam.**
- **35% Final exam.**

¹[Simon and Blume \(1994\)](#) and [Chiang and Wainwright \(2005\)](#) are other textbooks widely used in Economics. Either can serve as a useful reference.

Schedule of Class

Date	Reading/Topic
Sep. 1	1. Introduction to Analytical Tools
Sep. 3	2. Functions and Economic Relationships
Sep. 8	3. Limits, Continuity, and Tangent Lines
Sep. 10	4. Derivatives and Differentiation Rules
Sep. 15	5. Differentials and Basic Applications of Derivatives
Sep. 17	6. Second Derivatives, Concavity, and Convexity
Sep. 22	7. Optimization: Finding Maxima and Minima
Sep. 24	Review
Sep. 29	Midterm Exam
Oct. 1	8. Linear Algebra: Vectors and Matrices
Oct. 6	9. Linear Algebra: Solving Systems and Determinants
Oct. 8	10. Multivariable Functions: Partial Derivatives and the Total Differential
Oct. 13	11. Unconstrained Optimization: First-Order Conditions
Oct. 15	12. Unconstrained Optimization: Second-Order Conditions and the Hessian
Oct. 20	13. Constrained Optimization: The Lagrangian Method
Oct. 22	14. Constrained Optimization: Applications
Oct. 29	15. Comparative Statics and the Implicit Function Theorem
Nov. 3	16. The Envelope Theorem
Nov. 5	17. Integration
Nov. 10	Review
TBD	Final Exam

Standardized Information for All Courses

Attendance

- Attendance is mandatory. Punctuality and attendance will be noted.
- Each student may miss 10% of classes for any reason, with no excuse needed, and without penalty: 1, 2, or 3 classes, respectively, in a 1.5, 3, or 4.5 credit course. Each additional absence (for any reason) will result in a 4%, 6%, or 12% final grade penalty for a 4.5, 3.0, or 1.5 credit course, respectively.
- Tardiness of 20 minutes counts as an unapproved absence.
- Missing more than 25% of the classes in a course (without medical excuse), including “free” absences, will result in failing the course.
- Medical absences must be approved by the University Dean and communicated by the Dean to the instructor.

Accessibility Statement

Please review the University Accessibility Statement in the student catalog. Students having special needs should contact the Polaris Center or email Accomodations@uaustin.org.

Disability Support Services: The university will make reasonable accommodations for students with disabilities in compliance with Section 504 of the Rehabilitation Act and the Americans with Disabilities Act. The purpose of accommodations is to provide equal access to educational opportunities for eligible students with academic and/or physical disabilities.

Electronics

Please do not use laptops, iPads, cell phones, or any other electronic devices during class without explicit permission from the instructor. I will allow iPads/Tablets for note-taking, provided they lie flat on your table at all times and you use a pen to write (not type) your notes. I reserve the right to revoke this policy at any time.

Academic Misconduct

Instructors at UATX have the authority to assess possible plagiarism, unauthorized use of artificial intelligence, and other forms of cheating in their courses. Normally, cheating will result in failing the assignment.

References

- Chiang, A. C. and Wainwright, K. (2005). *Fundamental Methods of Mathematical Economics*. McGraw-Hill, New York, 4th edition.
- Hoy, M., Livernois, J., McKenna, C., Rees, R., and Stengos, T. (2022). *Mathematics for Economics*. MIT Press, 4th edition.
- Simon, C. P. and Blume, L. (1994). *Mathematics for Economists*. W.W. Norton & Company, New York.